

Integrals

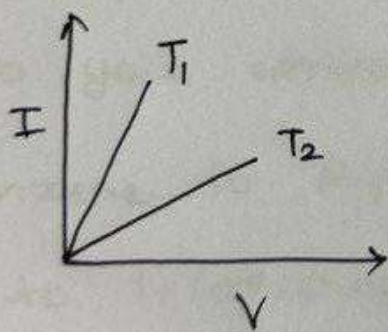
① $\int e^{2x} \cdot \sin(3x+1) dx$

② $\int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$

③ $\int \frac{(3 \sin \theta - 2) \cos \theta}{5 - \cos^2 \theta - 4 \sin \theta} d\theta$

④ Define the terms 'depletion layer' and 'barrier potential' for a p-n junction.

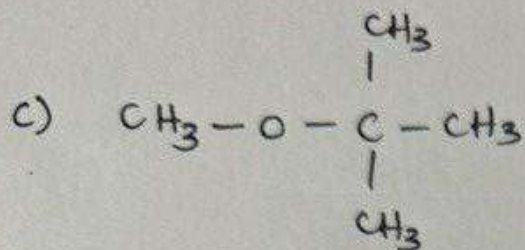
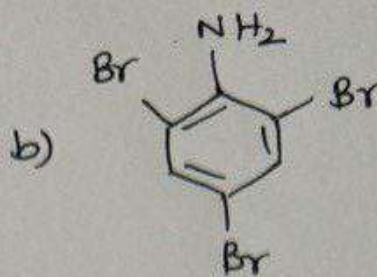
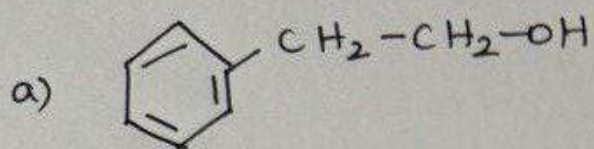
⑤ The V - I graph for a metallic wire at two different temperatures T_1 and T_2 is as shown in the figure. Which of the two temperatures is higher?



⑥ In what way is diffraction from each slit related to the interference pattern in a double slit experiment?

⑦ State the reasons why two independent sources of light cannot be considered coherent sources?

8) Give the IUPAC Names of given compound.



9) Write the chemical reactions involved in the following

① Hoffmann-bromamide degradation reaction.

② Carbylamine reaction.

10) Write the chemical reaction involved in Wolff-Kishner reduction?

11) How do you convert?

a) Chlorobenzene to Biphenyl

b) Propene to 1-iodopropane

c) 2-Bromobutane to but-2-ene.